

Team Worksheet — Lab 19

ANOVA, Interactions and Non-Parametric Tests (Black Belt Only)

Team members: _____

Date: _____

1. Our understanding of the problem

2. Our working

Use the data in `lab-19-anova-interactions-nonparametric.xlsx`.

3. What we produced

Deliverable: One-way and two-way ANOVA on your capstone data plus a non-parametric confirmation.

4. What the evidence does NOT tell us

5. Our conclusion

6. Step notes

Step 1. Understand alpha inflation: comparing 4 groups pairwise needs 6 t-tests, and at $\alpha = 0.05$ the family-wise false-positive risk rises to about 26 percent.

Step 2. Recap one-way ANOVA: it partitions total variation into between-group and within-group components, and the F-statistic is their ratio.

Step 3. State the ANOVA hypotheses: H_0 is that all group means are equal; H_a is that at least one differs — ANOVA does not tell you which.

Step 4. Check the ANOVA assumptions: normality of residuals, independence, and equal variances (test with Levene's or Bartlett's).

Step 5. Run one-way ANOVA on a capstone factor with 3+ levels — plant, line, shift, product family or supplier.

Step 6. If ANOVA is significant, run a post-hoc Tukey HSD to identify WHICH pairs differ, with the family-wise error rate controlled.

Step 7. Extend to two-way ANOVA with two factors at once — for example plant AND shift on your capstone Y.

Step 8. Read the interaction term. A significant interaction means the effect of one factor DEPENDS on the level of the other — for instance night shift hurts Suzhou but not Singapore.

Step 9. Plot the interaction. Non-parallel lines indicate interaction; crossing lines indicate a strong one that invalidates any single-factor conclusion.

Step 10. Understand why this matters for Improve: with a significant interaction, a fix that works at one plant may fail or backfire at another.

Step 11. Now the non-parametric branch. When normality fails and transformation does not fix it, switch tests rather than ignoring the violation.

Step 12. Map each test to its non-parametric equivalent: 2-sample t -> Mann-Whitney; one-way ANOVA -> Kruskal-Wallis; paired t -> Wilcoxon signed-rank; and Mood's median for heavy outliers.

Step 13. Re-run one of your significant findings using its non-parametric equivalent. Agreement across both strengthens your Day 5 defence considerably.

7. Self-check

Your ANOVA assumptions are tested and documented, significant results carry a post-hoc analysis, you have interpreted an interaction plot, and at least one finding is confirmed non-parametrically.

Complete? ☐ Yes ☐ Not yet

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